

SAI KIRAN VULLENGALA

Data Engineering | Software Engineering

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SUMMARY

Third-year B.Tech Data Science & Engineering student at IIT Palakkad who enjoys solving real-world problems in cross-functional teams. Led a 4-member engineering team shipping production features, and built full-stack, database-backed systems end-to-end with React, Node.js, and PostgreSQL. Strong foundation in DBMS, Data Structures & Algorithms, and Object-Oriented Programming, with 270+ LeetCode problems solved in C++. Comfortable collaborating with engineers and stakeholders across code reviews, sprint planning, and test-driven debugging.

EXPERIENCE

SDE Intern

June 2026 — July 2026

Vehant Technologies

India

- Built and debugged a Node.js and Python data pipeline to parse proprietary `.dump.pgm` and `.tip` X-ray formats, extract dual-energy pixel data, and generate 16-bit PNG and NumPy outputs for downstream processing.
- Developed a full-stack synthetic X-ray scan generation system using React, TypeScript, Node.js, Express.js, and Docker, supporting bulk ingestion, library management, interactive bag-region editing, and automated scan generation.
- Engineered a worker-pool processing system with batched task dispatch, concurrency control, polygon-based collision detection, and automated threat placement, reducing process-spawn overhead by up to 90% and enabling batch generation of synthetic scan layouts.

Technical Team Lead

May 2026 — Present

[IAR Cell, IIT Palakkad](#)

India

- Led a 4-member engineering team developing production features for the IIT Palakkad IAR Cell website, serving 2,000+ alumni records across public and authenticated portals through weekly sprints and code reviews.
- Built a React/Vite admin dashboard with full CRUD, global search, multi-field filtering, pagination, bulk operations, and CSV/Excel/PDF exports for alumni data management.
- Developed Express + Firebase REST APIs with token-based authentication, role-based access, duplicate-checked registration, profile management, and cursor-based pagination across 5+ endpoints.

EDUCATION

B.Tech in Data Science & Engineering, Indian Institute of Technology, Palakkad

Aug 2024 — Present

CGPA: 8.2/10

Palakkad, India

- Relevant Coursework: Data Structures & Algorithms, DBMS (PostgreSQL), Object-Oriented Programming, Artificial Intelligence, Optimization, Computer Systems

SKILLS

Programming Languages: Python, C++, SQL, JavaScript, TypeScript

Data & Databases: PostgreSQL, Supabase, DBMS, Data Pipelines, Database Design, REST APIs

Frameworks & Tools: React.js, Node.js, Git, GitHub, Docker, Vercel

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), Debugging & Testing, Agile Development, CI/CD

AI: Gemini API, Prompt Engineering, LLM Integration

PROJECTS

Aarogya HMS ([GitHub](#))

- Built a Hospital Management System using React 19, Node.js, and PostgreSQL with 4 role-based portals: Admin, Doctor, Nurse, and Patient, backed by a 15-table normalized, ER-modeled, and indexed relational schema designed to support scaling across multiple hospital departments.
- Automated billing and invoicing using PostgreSQL triggers; developed and tested 15+ API routes with Postman covering authentication, appointments, billing, and room management.

Nescafe Online Ordering System, IIT Palakkad ([GitHub](#))

- Developed a full-stack online ordering platform using React, Node.js, and Supabase to digitize order placement for the IIT Palakkad campus Nescafe outlet.
- Built a real-time Admin Command Center using Supabase Realtime for live order tracking, delivery batching, and inventory management.

ACHIEVEMENTS & HACKATHONS

- **LeetCode:** 270+ problems solved in C++ across Data Structures, Algorithms, and Problem Solving
- **Agentathon 2025, Hyderabad** – India's Largest Agentic AI Hackathon: Built **FinVoice** ([GitHub](#)), a multilingual voice banking assistant using the **Gemini Live API**, React 19, and TypeScript; enabled voice-driven transactions in English, Hindi, and Telugu via prompt-engineered conversational flows.